

CSA0404 – OPERATING SYSTEM

**OPERATING SYSTEM APPLICATIONS
AND SYSTEM CALLS**

PRESENTED BY :

PRASANNA A

192525130

BANKING SYSTEM

Operating System for Banking

- Banks process thousands of transactions every day.
The OS ensures secure, fast, and reliable processing.

Suitable OS Types

- Multi-user OS
- Multiprocessing OS
- Distributed OS

OS Structure

- User Applications
- System Call Interface
- Kernel
- Hardware

System Calls

- open()
- read()
- write()
- fork()

LINUX-BASED MULTI-USER SERVER

Resource Management in Linux

- Linux is widely used as a multi-user operating system in organizations for file sharing and device management.

Operating System Functions

- Process Management
- CPU Scheduling
- Memory Management
- File Management
- Device Management
- Security Management

User Process and Kernel Interaction

- 1.User requests a service.
- 2.System call is generated.
- 3.Kernel processes the request.
- 4.Hardware performs the operation.
- 5.Result is returned to the user.

REAL-TIME EMBEDDED SYSTEM FOR TRAFFIC SIGNAL CONTROL

Traffic Signal Control System

- Traffic signal controllers require immediate responses to control traffic safely.

Type of Operating System

- **Hard Real-Time Operating System (RTOS)**

Examples:

- FreeRTOS

Justification

- Fast response time
- Predictable execution
- High reliability
- Ensures road safety

Components

- Scheduler
- Timer
- Interrupt Handler
- Memory Manager

System Calls

- **sleep()** – Delays execution.
- **alarm()** – Generates timer signals.
- **timer_create()** – Creates timers for precise timing.

DISTRIBUTED COMPUTING SYSTEM

Distributed Operating System

- A distributed operating system allows multiple computers at different locations to work together as a single system.

Characteristics

- Resource Sharing
- Transparency
- Scalability

Architectaure

- User Applications
- Distributed Kernel
- Communication Layer
- Network

Communication System Calls

- socket()** – Creates communication endpoints.
- connect()** – Connects to another computer.
- send()** – Sends data.
- recv()** – Receives data.

Coordination System Calls

- fork()**
- wait()**
- pipe()**

OPERATING SYSTEM FUNCTIONS IN MODERN COMPUTERS

Key Operating System Functions

- An Operating System manages computer resources and provides services for application software.

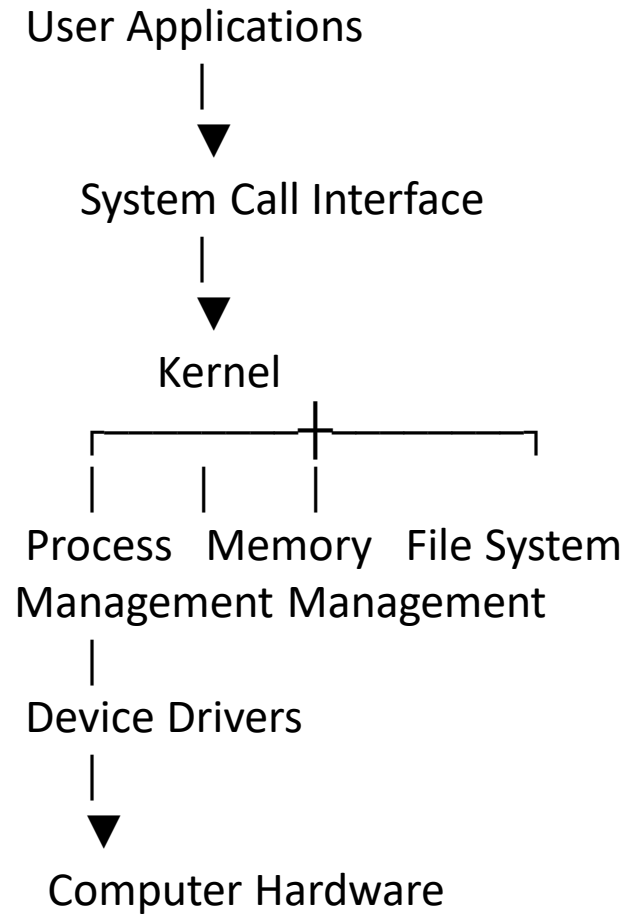
Main Functions

- Process Scheduling
- Memory Allocation
- File Management
- Device Management
- Input/Output Operations
- Security Management

Importance

- Improves system performance
- Efficient CPU utilization
- Protects user data
- Supports multitasking
- Provides reliable execution

COMMON OPERATING SYSTEM STRUCTURE



COMMON SYSTEM CALLS

Category	Examples	Purpose
Process Management	fork(), exec(), wait()	Create and control processes
File Management	open(), read(), write(), close()	Manage files
Memory Management	mmap(), brk()	Allocate memory
Device Management	ioctl(), mount()	Control devices
Communication	socket(), send(), recv(), connect()	Network communication
Timing	sleep(), alarm(), timer_create()	Time management

COMPARISON OF DIFFERENT OPERATING SYSTEMS

Scenario	Suitable Operating System	Main Feature
Banking System	Multi-user + Multiprocessing OS	High-speed transaction processing
Linux Server	Multi-user Linux	Resource sharing
Traffic Signal	Hard Real-Time OS	Immediate response
Distributed Computing	Distributed OS	Network-based resource sharing
Modern Computer	General Purpose OS	Multitasking and resource management
Scenario	Suitable Operating System	Main Feature

CONCLUSION

- Real-Time Operating Systems provide predictable and fast responses for critical applications such as traffic signal control.
- Distributed Operating Systems enable multiple computers to work together across different locations.
- System calls act as the communication bridge between user applications and the operating system kernel.
- An efficient operating system improves performance, security, reliability, and overall system efficiency.

THANK YOU